

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Sanjay kumar soy certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

sanjay kumar soy

Signature of the student:

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

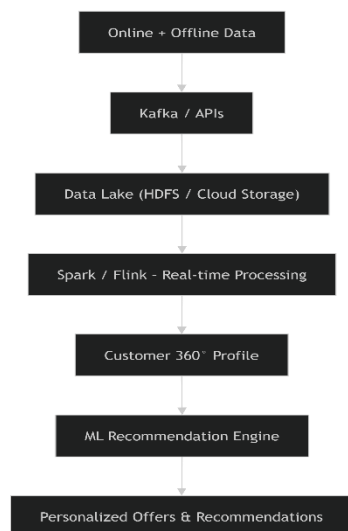
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

Answer

1. **Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.**



- Kafka/APIs collect data from different channels in real time.
- Data Lake stores customer transactions and behavior data.
- Spark/Flink processes the data and creates a unified Customer 360° profile.
- Machine Learning recommendation models such as collaborative filtering analyze customer behavior and provide personalized products, offers, and advertisements.

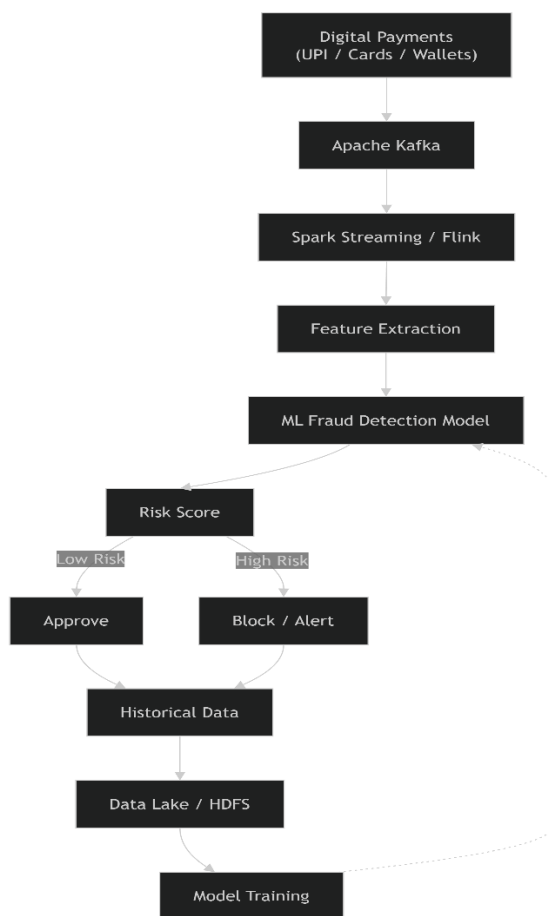
2. **Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.**

Distributed computing is **highly effective** for processing large-scale scientific data because it divides a large task among multiple computers instead of depending on one centralized system.

- **Parallel processing:** Large datasets are divided into smaller tasks and processed simultaneously by multiple nodes, reducing processing time.
- **Scalability:** More computers can be added to handle increasing data volumes, making the system more scalable than a centralized system.
- **Efficiency and reliability:** Workloads are distributed, so computational resources are used efficiently, and failure of one node does not necessarily stop the entire process.

Example: Scientific applications such as climate modeling, genomic analysis, and astronomical data processing can distribute calculations across hundreds or thousands of machines. Distributed and parallel computing provides faster processing, better resource utilization, scalability, and higher reliability compared with centralized systems.

3.Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.



4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management

Multimedia data such as **videos, audio, and images** creates challenges because it is very large, unstructured, and generated at high speed.

Challenges:

- Requires huge storage capacity due to large file sizes.
- Processing and transferring multimedia data requires **high computational power and bandwidth**.
- Data has different formats and structures, making integration and analysis difficult.
- Real-time processing of video and audio can cause **high latency**.

Suitable Technologies:

- HDFS – distributes large multimedia files across multiple nodes.
- Cloud storage (Amazon S3, Azure Blob Storage) – provides scalable and cost-effective storage.
- Apache Spark – enables distributed processing and analysis of multimedia datasets.
- NoSQL databases (MongoDB, Cassandra) – useful for storing multimedia metadata and associated information.
- Compression and formats such as JPEG, MP4 and AAC can reduce storage and bandwidth requirements.

5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

A cloud-based big data platform should use a **disaster recovery (DR) and business continuity plan** to protect data and keep services available during hardware failures, cyberattacks, network failures, or natural disasters.

Primary Cloud Region



Data Replication



Secondary Cloud Region



Automated Failover



Backup System / Recovery

- **Redundancy:** Deploy multiple servers/nodes and availability zones so that failure of one component does not stop the system.
- **Replication:** Maintain copies of important data in different servers or cloud regions. Technologies such as **HDFS replication, cloud storage replication, and database replication** can be used.
- **Regular Backups:** Take scheduled backups and store them separately to recover from data loss or corruption.
- **Automated Failover:** Use cloud load balancers and health checks to automatically redirect users to a healthy server or secondary region when the primary system fails.
- **Business Continuity:** Define **RTO (Recovery Time Objective)** and **RPO (Recovery Point Objective)** to determine how quickly services must be restored and how much data loss is acceptable.

6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Data quality problems such as inconsistency, duplication, and missing values can seriously affect the reliability of large-scale analytics.

- **Inconsistency:** Different systems may store the same data in different formats or values. This can produce incorrect analysis and misleading results.
- **Duplication:** Repeated records can lead to **over-counting**, incorrect statistics, and inaccurate customer or transaction analysis.
- **Missing values:** Incomplete data can reduce the accuracy of analytics and machine learning models and may lead to incorrect predictions.

Data Governance Strategies

- Establish data quality standards for accuracy, consistency, completeness, and validity.
- Use data cleansing and validation tools to identify and correct errors.
- Apply deduplication techniques to remove repeated records.
- Handle missing values using appropriate methods such as imputation or predefined defaults.
- Maintain a data catalog and metadata to track data sources, owners, and definitions.
- Assign data owners/stewards to monitor data quality.
- Perform regular data quality audits and monitoring.

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement



- ❖ Kafka collects user activities such as viewing history, searches, likes, ratings, and watch time in real time.
- ❖ Spark/Flink performs distributed processing of large volumes of user behavior data.
- ❖ Collaborative filtering recommends content based on similarities between users and their viewing patterns.
- ❖ Machine learning can analyze factors such as genre preferences, watch time, ratings, and previous interactions to improve recommendations.
- ❖ Distributed processing allows millions of users and interactions to be processed simultaneously, providing scalability and faster recommendations.
- ❖ Continuous analysis of user behavior improves recommendation accuracy, while personalized content increases user engagement, watch time, and customer retention.

8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Advantages:

- **Scalability:** Resources such as storage and processing power can be increased or decreased according to the volume of data.
- **Cost Management:** Organizations follow a pay-as-you-go model, reducing the need for expensive hardware and maintenance.
- **Flexibility:** Cloud platforms provide various services for data storage, processing, analytics, and machine learning that can be accessed from anywhere.
- **High Availability:** Data can be replicated across multiple servers or regions, improving reliability and disaster recovery.

Limitations:

- **Security and Privacy:** Sensitive data stored in the cloud may face unauthorized access, data breaches, or compliance issues.
- **Cost Overruns:** Continuous use of large-scale cloud resources can become expensive if not properly monitored.
- **Vendor Lock-in:** Dependence on one cloud provider can make migration to another platform difficult.
- **Internet Dependency:** Cloud-based big data applications require reliable network connectivity.

9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

- Apache Kafka collects real-time data from GPS devices, vehicles, traffic systems, and IoT sensors.
- Spark Streaming or Apache Flink processes the incoming data continuously and detects traffic congestion, delays, and changes in vehicle conditions.
- Route optimization algorithms such as Dijkstra's algorithm or A* can identify the shortest or fastest route using real-time traffic information.
- Machine learning models can predict traffic, travel time, fuel consumption, and possible delivery delays.
- Historical and real-time data helps managers make decisions about vehicle allocation, delivery schedules, maintenance, and route changes.

Benefits:

- Route optimization: Avoids congested roads and reduces travel time.
- Fuel efficiency: Selects shorter routes and identifies inefficient driving patterns.
- Better decisions: Provides real-time information for fleet managers.
- Reduced delays: Predicts traffic and delivery problems before they occur.



10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Challenges:

- Privacy: Collecting personal information without proper consent can violate user privacy.
- Data Security: Large databases can become targets for hacking, data breaches, and unauthorized access.
- Lack of Transparency: Users may not know how their data is collected, analyzed, or shared.
- Bias and Discrimination: Biased datasets or algorithms can produce unfair decisions.
- Data Ownership: Organizations must clearly determine who owns and controls collected data.
- Legal Compliance: Organizations must follow applicable data-protection and privacy regulations.

Recommended Organizational Policies

- Obtain clear user consent before collecting personal data.
- Follow data minimization—collect only necessary information.
- Use encryption, authentication, and access controls to protect data.
- Anonymize or pseudonymize personal information wherever possible.
- Conduct regular privacy and security audits.
- Maintain clear data retention and deletion policies.
- Monitor AI/ML models for bias and unfair outcomes.
- Provide users with transparency about how their data is used and shared.
- Train employees on data privacy, security, and ethical data handling.